ARTIN APPROXIMATION OF A LOCAL MAP FROM A SCHEME INTO ANOTHER BY AN ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11016-11039 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11041-11068, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. G-rings, completions of local rings, elementary etale neighbourhoods, and Artin approximation; the construction is retained.

Source reference. Original lemma 07PV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11068–11068. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CAR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11068–11068. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let X, Y be schemes locally of finite type over S. Let $x \in X$ and $y \in Y$ be points lying over the same point $s \in S$. Assume $\mathcal{O}_{S,s}$ is a G-ring. Assume further we are given a local $\mathcal{O}_{S,s}$-algebra map*

$$\varphi : \mathcal{O}_{Y,y} \longrightarrow \mathcal{O}^\wedge_{X,x}$$

For every $N \geq 1$ there exists an elementary étale neighbourhood $(U, u) \to (X, x)$ and an S-morphism $f : U \to Y$ mapping u to y such that the diagram

$$\begin{array}{ccc} \mathcal{O}^\wedge_{X,x} & \longrightarrow & \mathcal{O}^\wedge_{U,u} \\ {\scriptstyle\varphi}\uparrow & & \uparrow \\ \mathcal{O}_{Y,y} & \xrightarrow{f_u^\sharp} & \mathcal{O}_{U,u} \end{array}$$

commutes modulo $\mathfrak{m}_u^N$.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The question is local on X hence we may assume X, Y, S are affine. Say $S = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$. Write $B = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to x. The local ring $\mathcal{O}_{X,x} = A_{\mathfrak{p}}$ is a G-ring by More on Algebra, Proposition 07PV. The map φ is a map

$$B^\wedge_{\mathfrak{q}} \longrightarrow A^\wedge_{\mathfrak{p}}$$

where $\mathfrak{q} \subset B$ is the prime corresponding to y. Let $a_1, \ldots, a_n \in A^\wedge_{\mathfrak{p}}$ be the images of $x_1, \ldots, x_n$ via $R[x_1, \ldots, x_n] \to B \to B^\wedge_{\mathfrak{q}} \to A^\wedge_{\mathfrak{p}}$. Then we can apply Smoothing Ring Maps, Lemma 0CAR to get an étale ring map $A \to A'$ and a prime ideal $\mathfrak{p}' \subset A'$ and $b_1, \ldots, b_n \in A'$ such that $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}')$, $a_i - b_i \in (\mathfrak{p}')^N (A'_{\mathfrak{p}'})^\wedge$, and $f_j(b_1, \ldots, b_n) = 0$ for $j = 1, \ldots, n$. This determines an R-algebra map $B \to A'$ by sending the class of x_i to $b_i \in A'$. This finishes the proof by taking $U = \mathrm{Spec}(A') \to \mathrm{Spec}(B)$ as the morphism f and $u = \mathfrak{p}'$. $\square$